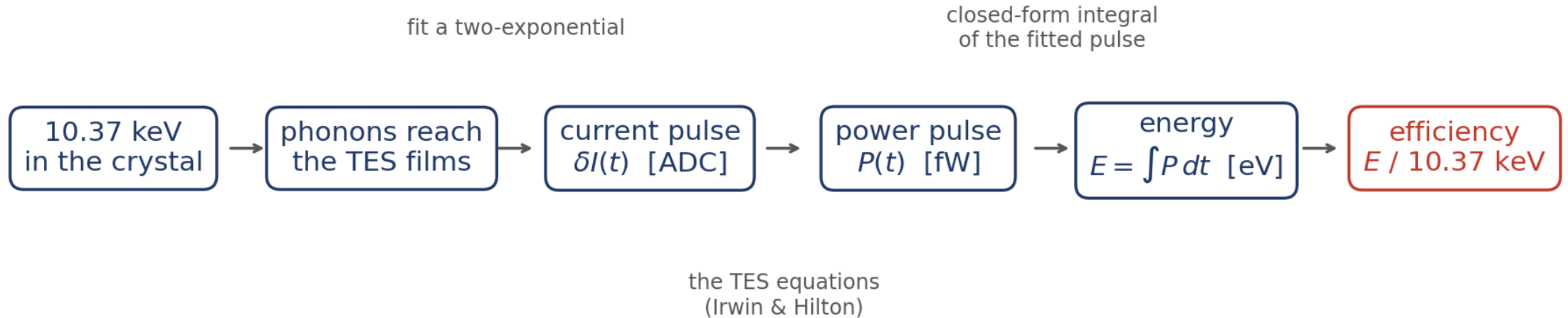


Phonon collection efficiency of Z7 from the 10.37 keV K line

the energy a K-line event leaves in the TESs, from the fitted current pulse through the TES equations

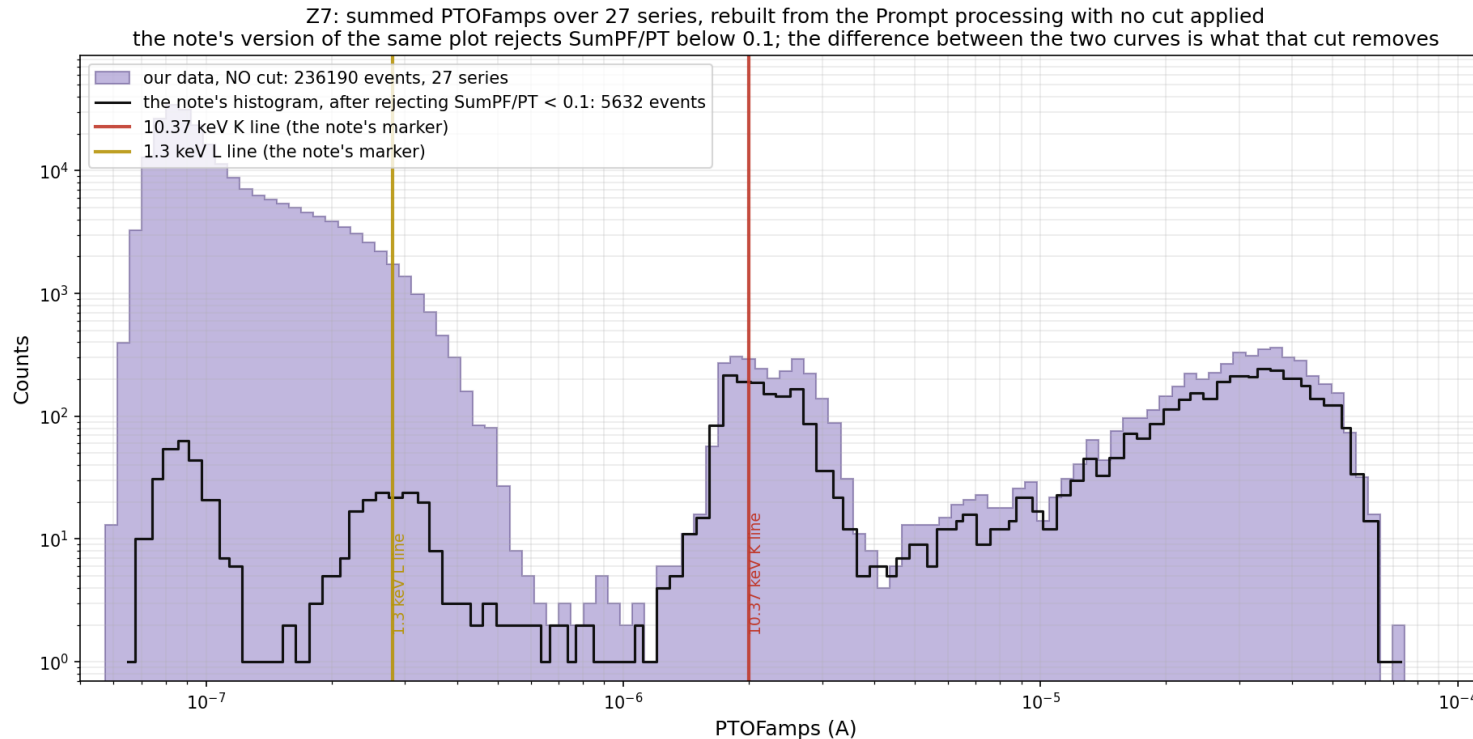
Zhiheng Li · University of Minnesota · SuperCDMS SNOLAB Run 4 · September 2026
github.com/ZhihengLi0/urop → [snolab/collection_efficiency/](https://github.com/ZhihengLi0/urop/tree/main/snolab/collection_efficiency/)

The question, and why the trace alone does not answer it



- Collection efficiency = the energy that ends up in the TES films ÷ the 10.37 keV the event deposited
- What we record is a current: the TES current changes by $\delta I(t)$ when phonons heat the film
- **The area under the current trace is a charge, not an energy — it cannot be compared with 10.37 keV**
- So the current pulse has to be turned into a power pulse first, using the TES equations; the area under the power pulse is the energy
- Everything today is one detector, Z7 (the quietest one), and the K-line events of the Ge activation run

The events: the 10.37 keV K line of the Ge activation run, on Z7



Summed PTOFamps of Z7 rebuilt from the Prompt processing (purple); black: the ops note's histogram, which rejects SumPF/PT < 0.1; red line: the K line.

- After the Cf activation every detector produces mono-energetic 10.37 keV events (71Ge electron capture)
- On Z7 the K line is well separated from the noise triggers: a clean sample of identical events
- 2207 K-line events over 30 series; for each one the raw trace of every phonon channel was saved (11 channels read out on Z7)
- For each channel the bias point (I_0 , R_0 , R_p , R_{sh}) is read from the processing's detectorConfig
- Same sample as the pulse-template work — nothing new was selected

From current to power: the TES equations, small-signal

$$P(t) = I_0 (R_L - R_0) \delta I(t) + 2 R_L (\delta I(t))^2$$

linear term

quadratic term

$$E = \int P(t) dt = I_0 (R_L - R_0) A (\tau_f - \tau_r) + 2 R_L A^2 \left(\frac{\tau_f + \tau_r}{2} - \frac{2 \tau_f \tau_r}{\tau_f + \tau_r} \right)$$

for the fitted pulse $\delta I(t) = A [e^{-t/\tau_f} - e^{-t/\tau_r}]$ the integral is exact, no window needed

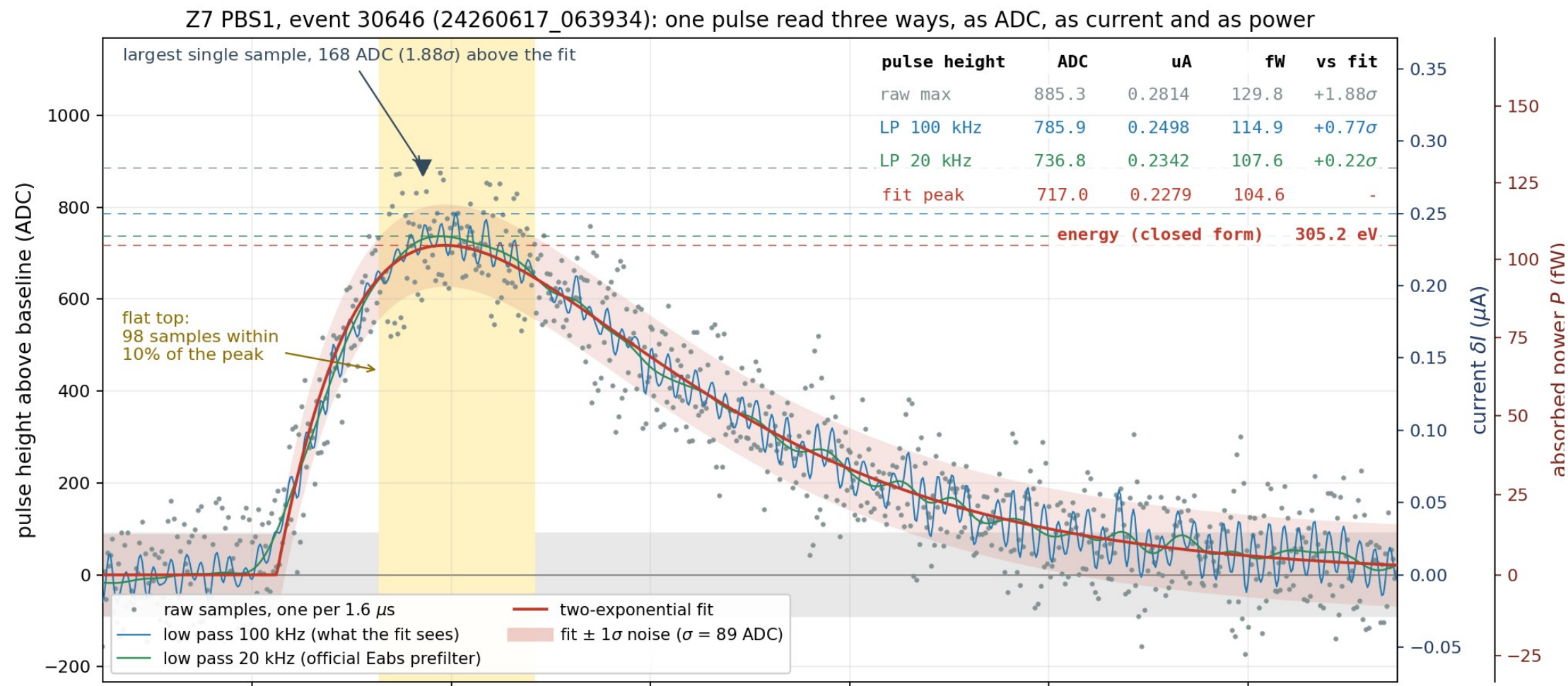
Z7 PBS1 bias point (measured, from detectorConfig): $I_0 = -18.07 \mu\text{A}$, $R_0 = 44.16 \text{ m}\Omega$, $R_L = R_p + R_{sh} = 19.24 \text{ m}\Omega$

$$\Rightarrow I_0 (R_L - R_0) = 4.50 \times 10^{-7} \text{ V} \quad \text{and} \quad 2 R_L = 0.0385 \Omega$$

- The power absorbed by the film is the change of Joule heating when the current moves by δI : a linear term plus a quadratic term (Irwin & Hilton; Method 1 of the CDMS collection-efficiency note)
- Both coefficients come from the measured bias point of that channel — nothing is fitted or tuned
- The trace is in ADC counts: $3\,145\,728\,000 \text{ ADC} = 1 \text{ A}$, so $1 \text{ ADC} = 0.318 \text{ nA}$. Only the change above the baseline is used
- Three variants of the formula exist ($c_2 = 2, 1, -1$); they differ by under 1.2 % in energy, so the choice is not the main uncertainty (backup)

One pulse read three ways: ADC, current, power — the height

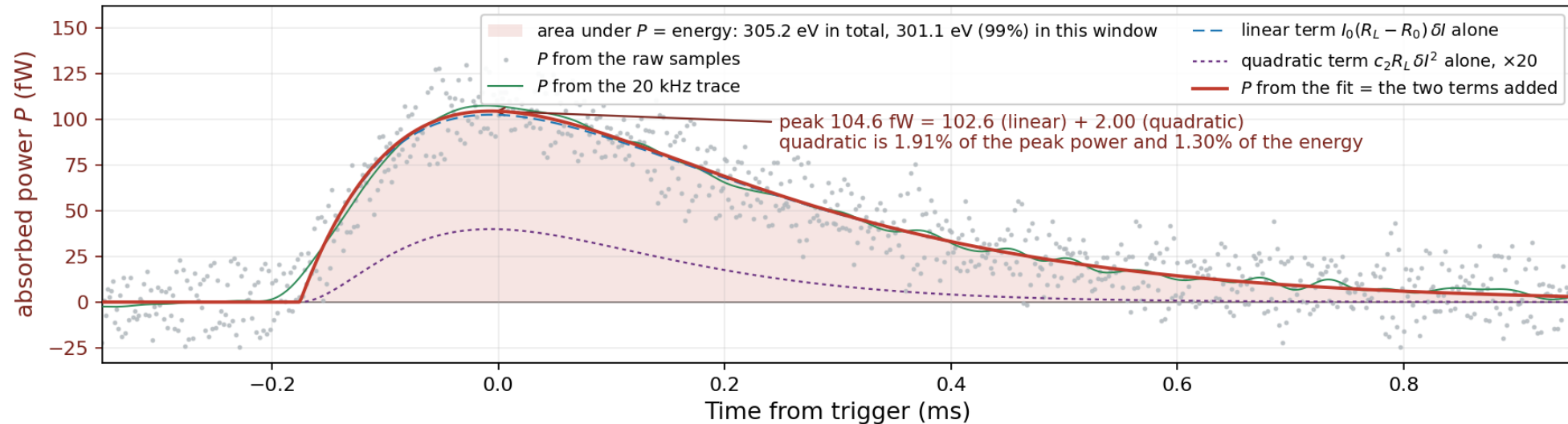
Z7 PBS1, event 30646, zoomed on the peak; every curve is the same trace



Grey: raw samples. Blue: 100 kHz low pass (what the fit sees). Green: 20 kHz (the official Eabs prefilter). Red: the two-exponential fit; pink band = fit ± 1σ noise. Yellow: the flat top, 98 samples within 10 % of the peak. Largest raw sample 885 ADC against 717 from the fit: +23 %.

One pulse read three ways — the power, and its area

same event; the fitted current put through the formula



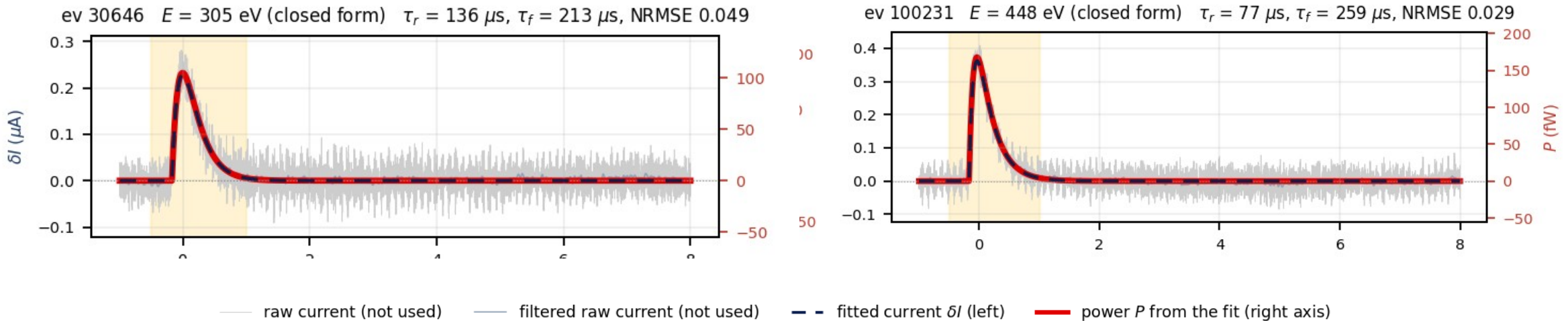
baseline = mean of bins 93-15758 = 28667.7 ADC, subtracted | 1 ADC = 0.3179 nA | power axis is exact, so its ticks are not evenly spaced

$P = I_0(R_L - R_0) \delta I + c_2 R_L \delta I^2$ (method1, $c_2=2$): $I_0=-18.069 \mu\text{A}$, $R_0=44.159 \text{ m}\Omega$, $R_L=19.240 \text{ m}\Omega \Rightarrow 4.5026\text{e-}07 \text{ V}$ and 0.03848Ω

- Red: the power pulse from the fitted current — the two terms added. Blue dashed: the linear term alone, almost the whole thing. Purple dotted: the quadratic term alone, drawn $\times 20$ so it can be seen
- **Peak 104.6 fW = 102.6 (linear) + 2.0 (quadratic): the quadratic term is 1.9 % of the peak power and 1.3 % of the energy — the power pulse has the shape of the current pulse**
- **The shaded area is the energy this channel absorbed: 305.2 eV; 99 % of it lies inside this 1.3 ms window**
- Grey dots and green: the same formula applied to the raw and 20 kHz traces, for comparison only. Bottom line: the measured bias point that fixes the two coefficients

Every event gets the same treatment: fit the current, convert to power

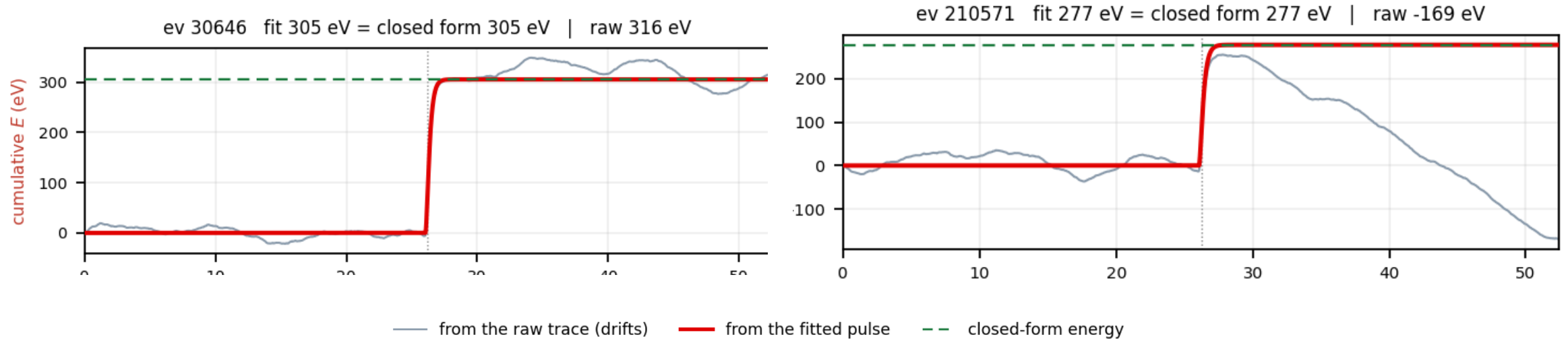
Z7 PBS1, two of the 15 example events of series 24260617_063934



- Each trace is fitted with $A \cdot [e^{(-t/\tau_f)} - e^{(-t/\tau_r)}]$ directly in current units, with the pulse start left free (± 4.8 ms around the trigger)
- Left axis: fitted current δI in μA (navy dashed). Right axis: the power it implies in fW (red). The raw trace (grey) is only drawn for reference
- The two curves lie on top of each other because the quadratic term is only 2 % — the right axis is just the left axis times 4.5×10^{-7} V
- Yellow band: the official Eabs integration window (-0.5 to $+1$ ms). The fit needs no window
- Fits are kept if the residual is under 20 % of the pulse height (NRMSE ≤ 0.2 ; typical values 0.03–0.07). Same channel, 15 events: energies from 198 to 448 eV

Why the fit is integrated and not the raw trace

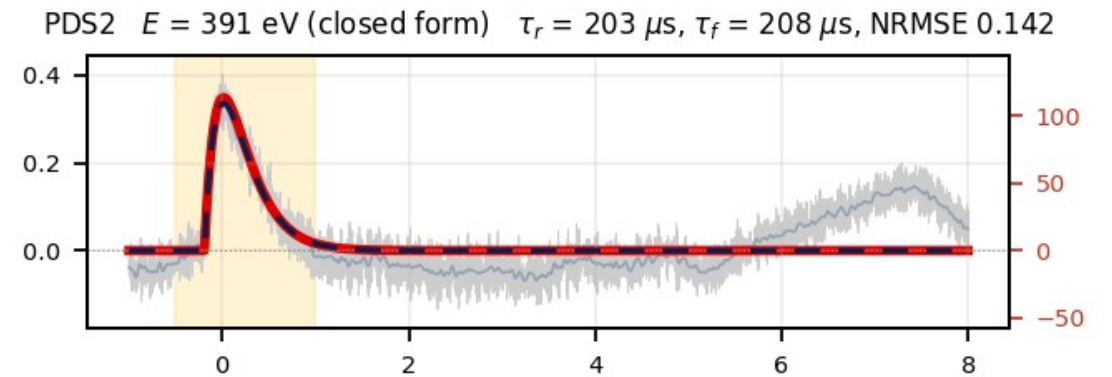
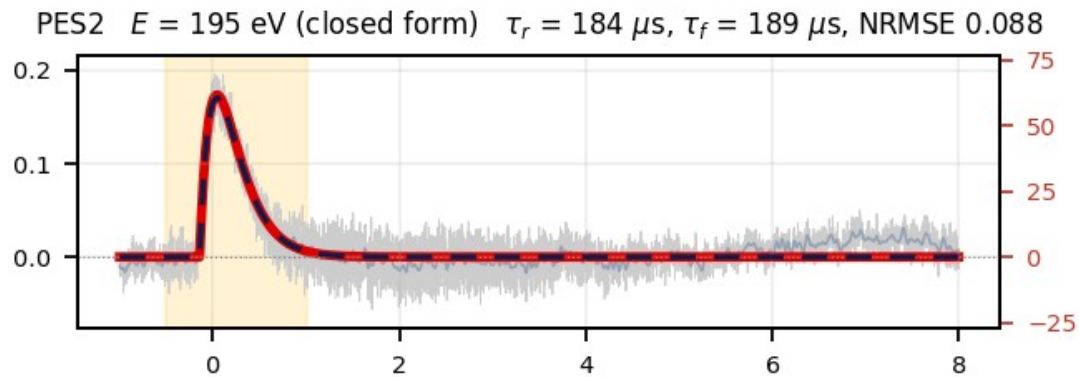
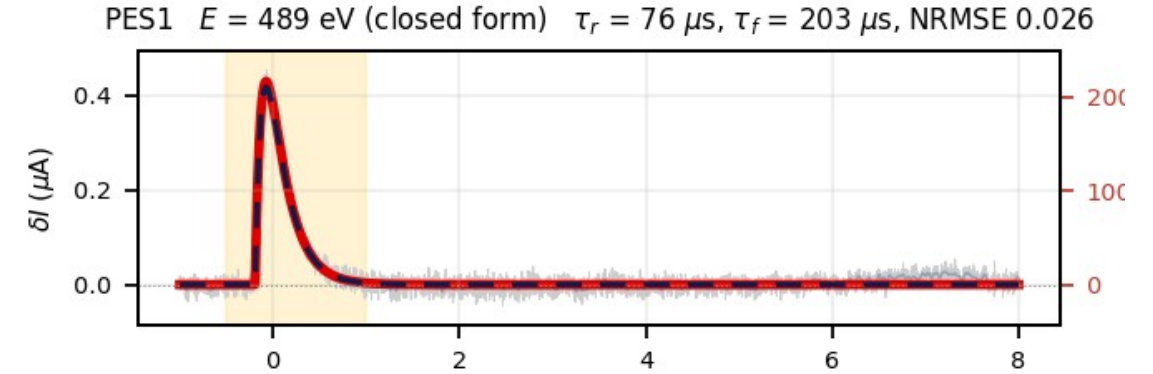
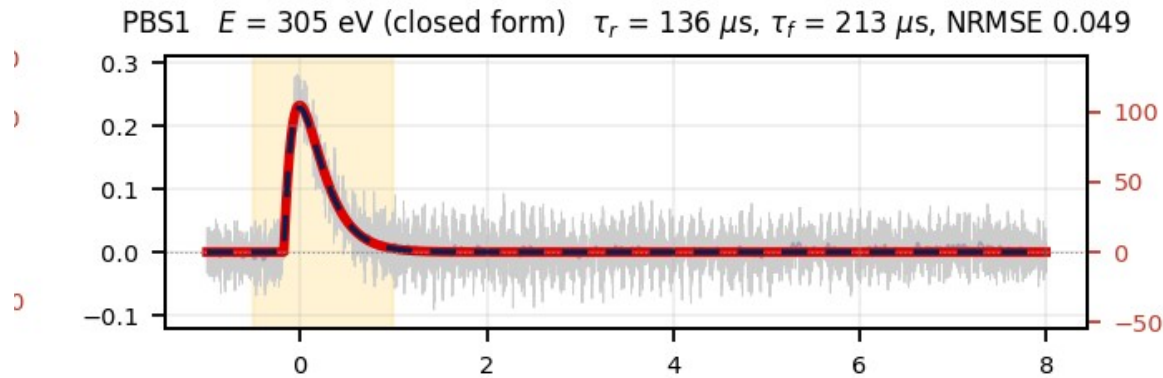
cumulative energy $\int_0^t P \, dt$ over the whole 52 ms trace, Z7 PBS1



- Horizontal axis: the whole trace, 0 to 52.43 ms; the trigger is the dotted line at 26.2 ms. Vertical: energy accumulated so far
- Red, from the fitted pulse: zero before the pulse, up within 1 ms, then flat. Its end value is exactly the closed-form integral (green dashed) — the energy does not depend on any window
- **Grey, from the raw trace: it drifts. A baseline offset of 1 nA over 15 ms already adds 42 eV. For the right-hand event the raw integral ends at -169 eV against 277 eV from the fit**
- This is the reason the energy is taken from the fitted pulse, as a formula in A , τ_r , τ_f , and not from a numerical integral of the data

One event, all channels: the energies add up to 33 % of 10.37 keV

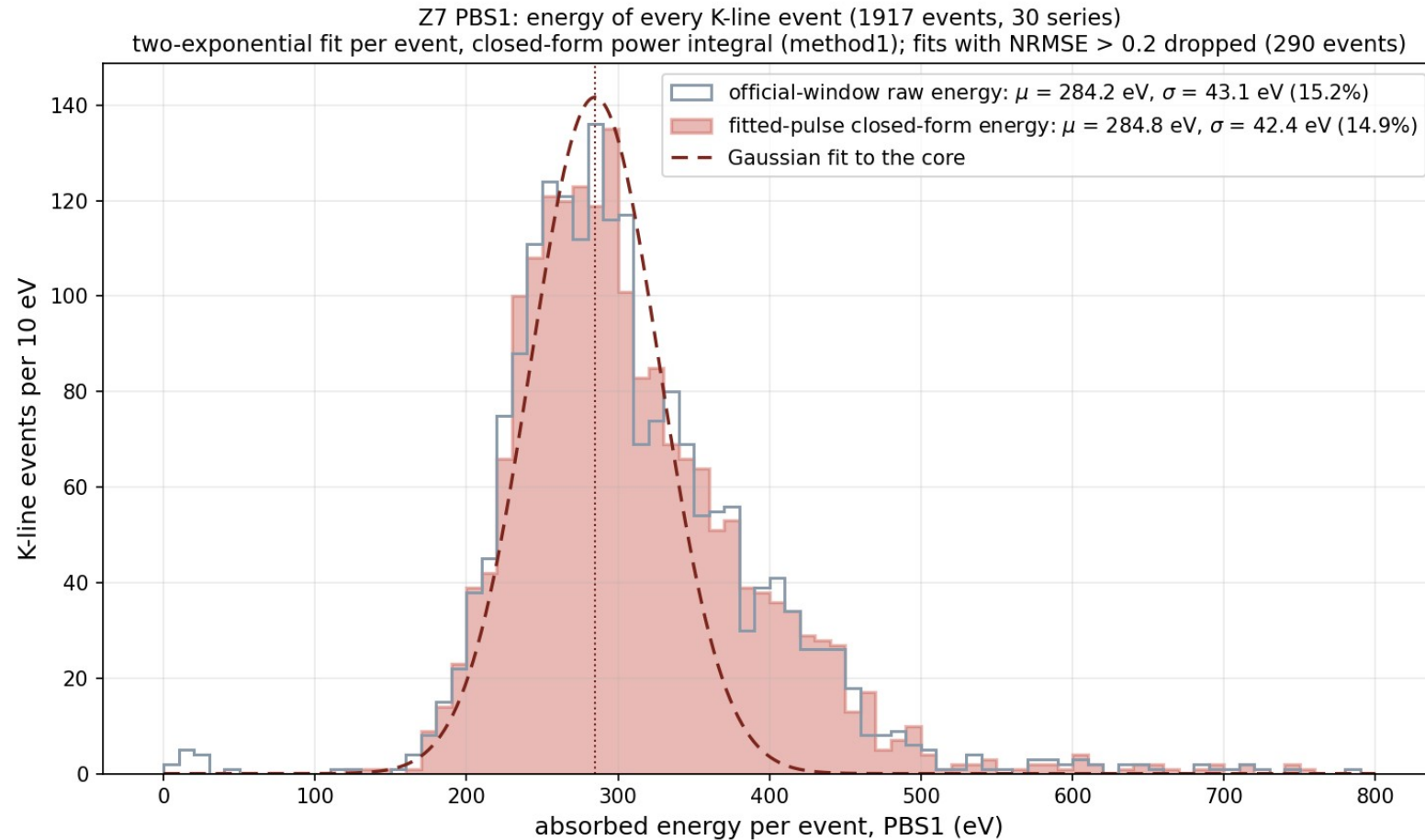
Z7 event 30646; four of the eleven channels shown, same axes as before



- The channels see very different shares: PES1 489 eV, PES2 195 eV — the event happened closer to one side. Time constants also differ (τ_f 150–250 μs)
- **Sum over the 11 channels: 3419 eV = 33.0 % of 10.37 keV. The official window applied to the raw traces gives 3397 eV = 32.8 %, 0.6 % apart**
- PDS2 (bottom right) carries a slow swing after 5 ms — a low-frequency artefact of that channel; its fit residual is 0.14, three times the others

Every K-line event, one channel: the energy distribution of PBS1

1917 events over 30 series; one entry = one event = one fit integrated



- Horizontal: energy absorbed in PBS1, 10 eV per bin. Vertical: number of events

- Red: from the fitted pulse. Grey outline: the official window on the raw trace. Same peak to 0.2 %

- **Peak 285 eV, width 42 eV = 15 % — for a line whose energy is fixed**

- The dashed curve is a Gaussian fitted to the core only, so μ is the peak position, not the mean

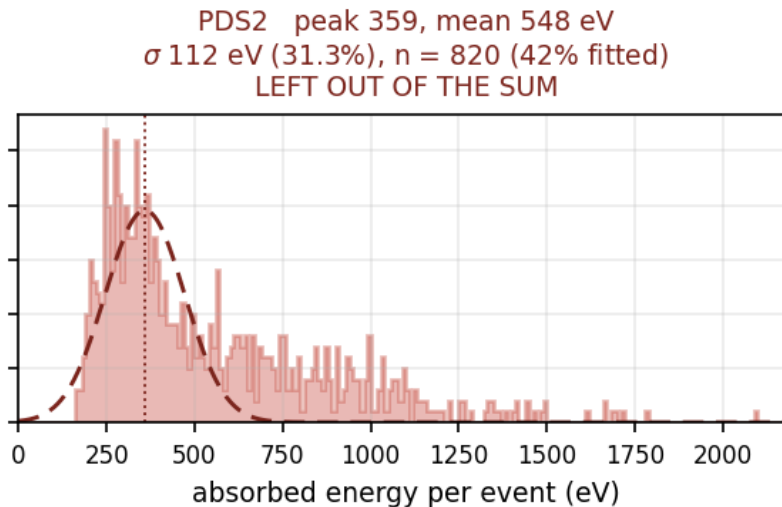
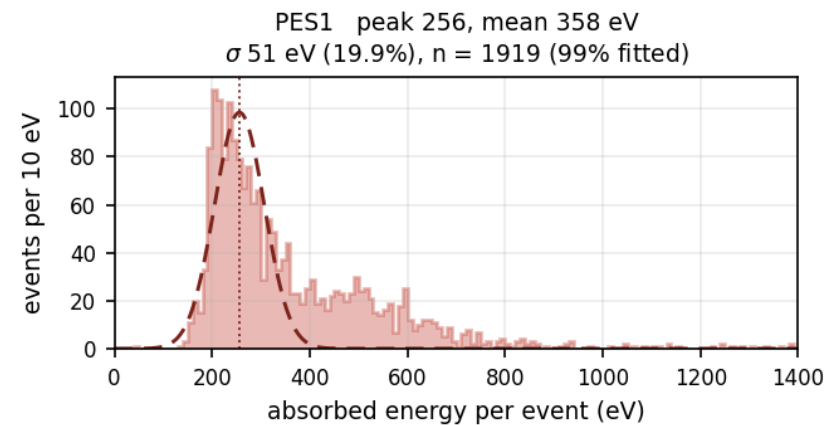
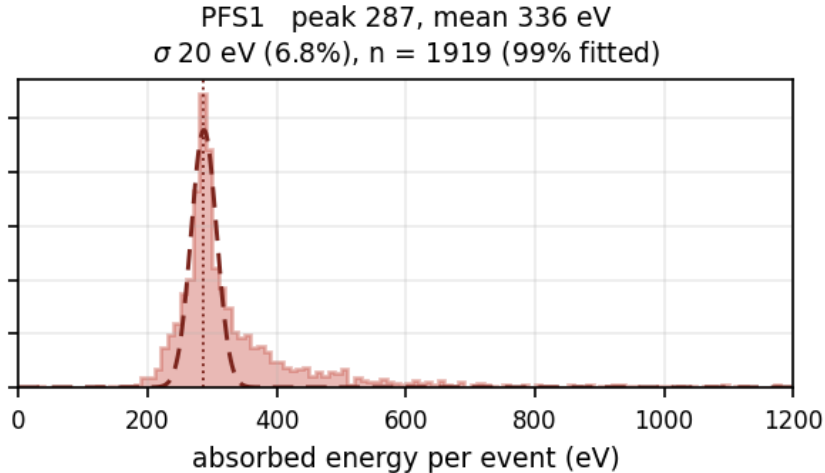
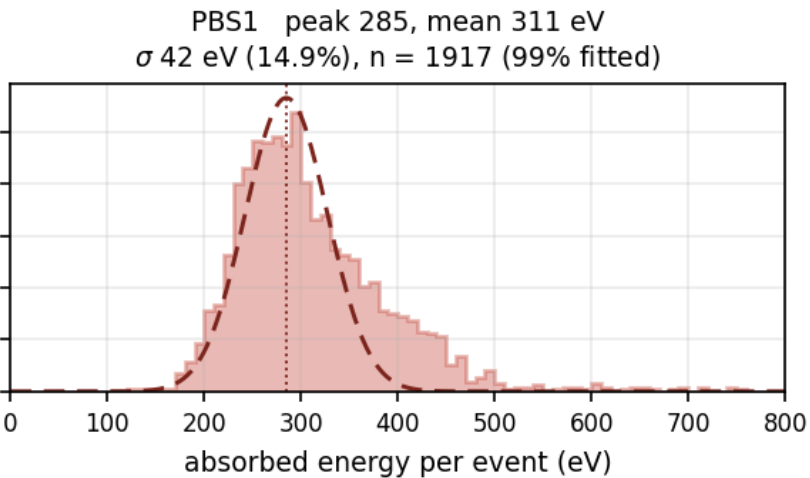
- The distribution is right-skewed: the tail to 600 eV is events that happened near this channel

- So one channel's width is mostly event position, not noise — the noise floor of a single fit is a few percent

- 290 of the 2207 events fail the fit-quality cut in this channel

Every channel: each is skewed, and the peaks must not be added

four of the eleven panels; all eleven in the backup



- Every channel is right-skewed, so its Gaussian peak sits below its mean: PBS1 285 vs 311 eV, PES1 256 vs 358 eV

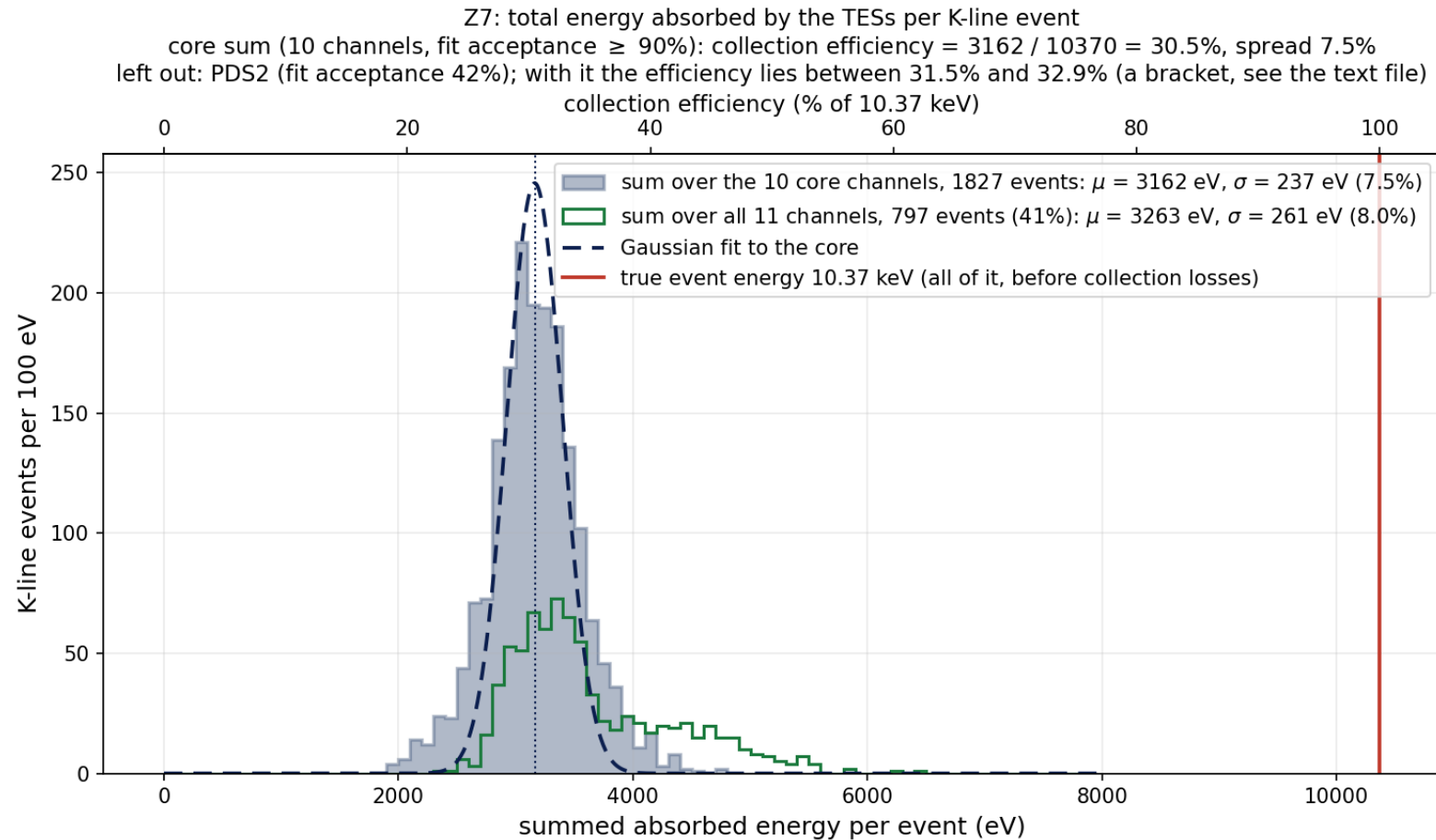
- Peaks of skewed distributions do not add — the sum of the eleven peaks is 21 % low. Means do add. The per-event sum on the next slide is the honest quantity

- Widths range from 6.8 % (PFS1) to 27 % (PDS1, PES2): the channels differ in how much of the position spread they see

- PDS2 fits only 42 % of the events (the low-frequency artefact fails the residual cut) and is left out of the core sum; ten channels remain, each fitting ≥ 95 % of events

Summed over channels: collection efficiency $32 \pm 1 \%$

the total energy the TESs absorbed per K-line event



- Bottom axis: the summed energy of one event; top axis: the same number as a fraction of 10.37 keV

- **Grey: the ten core channels, 1827 events — peak 3162 eV = 30.5 %, width 7.5 %**

- Summing over channels turns the skewed 15–27 % single-channel distributions into a symmetric 7.5 % one: the position dependence cancels

- Green: all eleven channels, but only the 797 events where PDS2 fits, 3263 eV. That subsample is biased — its ten-channel sum is 4.8 % low — so PDS2's share can only be bracketed

- **Full efficiency between 31.5 % and 32.9 %: $32 \pm 1 \%$**

- The way to shrink the bracket is to make PDS2 fittable, not more statistics

- Red line: 10.37 keV — about two thirds of the energy never reaches the TESs

Where 32 % sits, and what is not yet nailed down

- The CDMS collection-efficiency note (R37, CUTE tower) reports Z1 26.2 % / 29.2 % (0 V / 50 V), Z3 40.9 % / 41.2 %, Z6 26.3 % — our 32 % for Z7 falls inside that range
- Their method integrates the raw pulse until it has decayed by 90 %; ours integrates the fitted pulse in closed form. On the example event the official window and the fit agree to 0.6 %
- **Only Z7 so far. Every other detector needs its own bias points and its own fits — the scripts take the detector number**
- **The HV bias of Z7 during these series is not established; all numbers assume 0 V. With a bias, Luke phonons would enlarge the deposited energy and the denominator changes**
- **No dI/dV data on MSI, so loop gain and inductance are unknown; the small-signal formula drops the inductor term (a fast, small correction) — it could not be checked**
- PDS2's low-frequency artefact is the ± 1 %: fixing it collapses the bracket to one number
- 276 of the 2207 events (12.5 %) fit in no channel at all and are not in the histograms; whether they carry a different energy has not been checked

Summary

- The trace is a current. Turned into power with the TES equations (measured bias point per channel), its integral is the absorbed energy; the quadratic term is 2 %
- The energy is taken from the fitted pulse, not the data: no bias from the noisy maximum, no drift from the baseline, no integration window
- One channel: a 15 % wide, right-skewed distribution — event position. Summed over ten channels: symmetric, 7.5 % wide
- **Z7 absorbs 32 ± 1 % of a 10.37 keV event in its TESs; the ± 1 % is PDS2**
- Next: the other twelve detectors; make PDS2 fittable; settle the HV bias; get dI/dV for the loop gain and the inductor term; look at the 12.5 % of events that fit nowhere
- Code and figures: `snolab/collection_efficiency/` (scripts, fit cache, NOTES.md with the derivation)

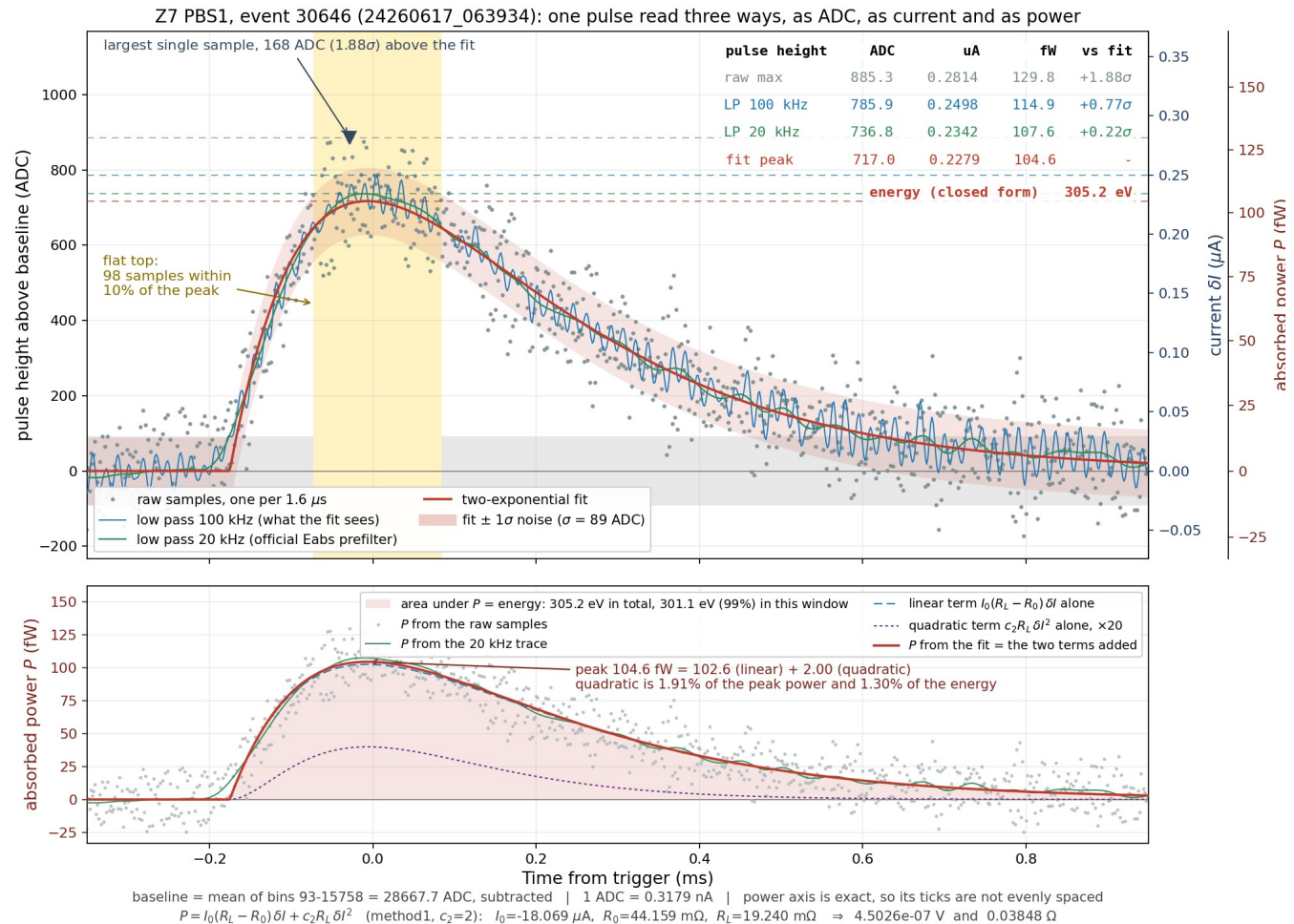
Backup — the three versions of the power formula

- $P = I_0(R_L - R_0) \cdot \delta I + c_2 \cdot R_L \cdot (\delta I)^2$
 - $c_2 = 2$: Method 1 of the CDMS collection-efficiency note — used throughout this talk
 - $c_2 = 1$: the exact small-signal expression from the two TES equations
 - $c_2 = -1$: Method 2 of the same note
- Because the quadratic term is 1.3 % of the energy, the three differ little: relative to the exact form, Method 1 is +0.58 % and Method 2 is -1.15 % on the example event
- The 15 % single-channel width and the PDS2 bracket are both far larger, so the formula choice is not the limiting uncertainty
- The inductor term of the full equations is neglected: it needs L and the loop gain from dI/dV , which do not exist for these series on MSI

Backup — the official Eabs, reproduced to 0.2 %

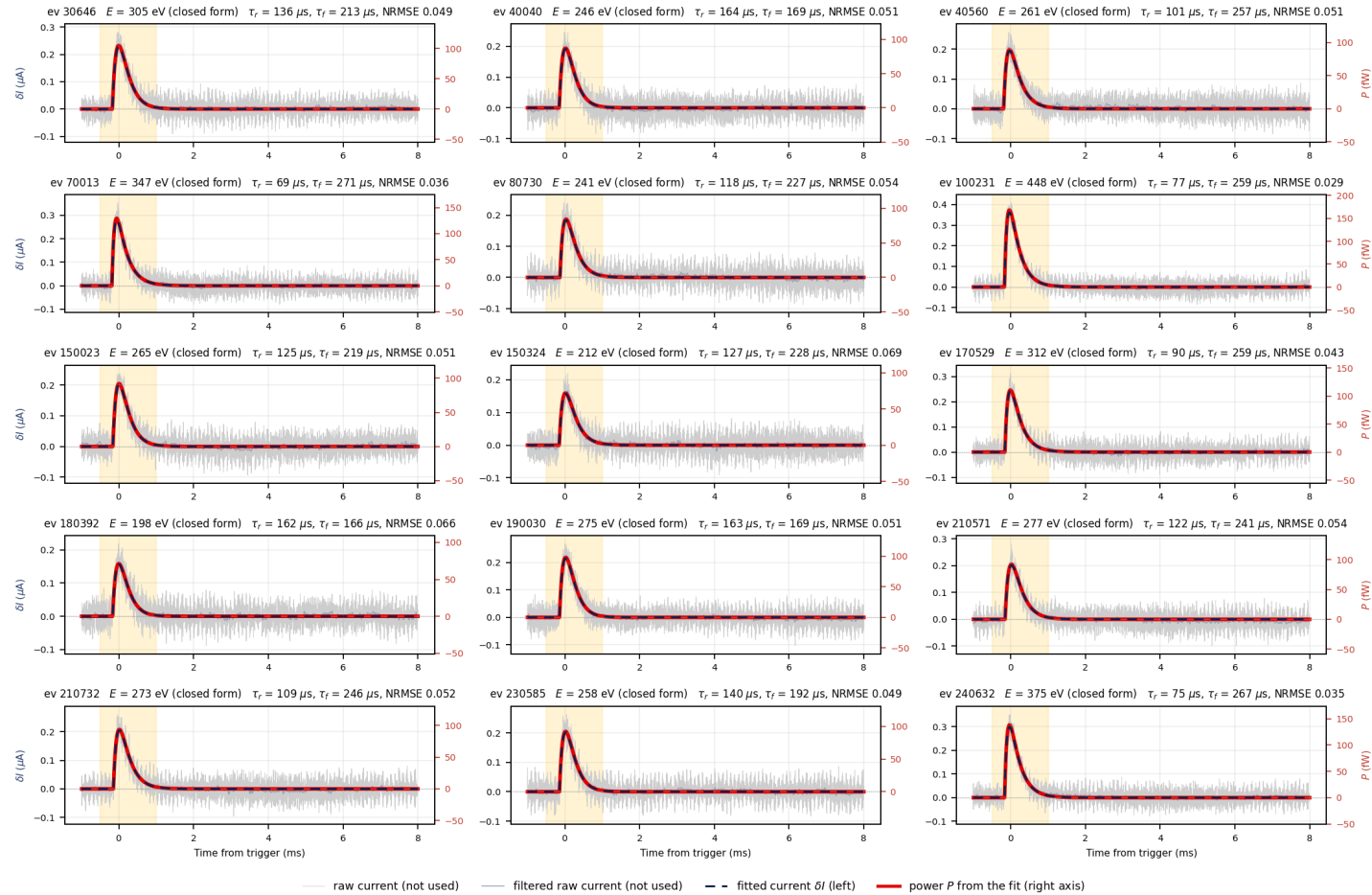
- Trigger at bin 16383 of 32768; sample spacing 1.6 μ s (52.43 ms per trace)
- Baseline = mean of bins 93 .. 15758, subtracted
- Prefilter: 5-pole 20 kHz Butterworth
- Window: -500μ s to $+1000 \mu$ s around the trigger (bins 16071 .. 17008)
- Power from the same Method-1 formula, summed over the window $\times 1.6 \mu$ s
- Our reproduction matches the official Eabs to 0.2 %; the grey outlines on the histogram slides are this quantity
- The fitted pulse puts 98–99 % of its energy inside that window on PBS1 (84–99 % over all channels, median 98.8 %) — the window is fine for the fit; what hurts the raw trace is the drift, not the tail

Backup — the full peak figure



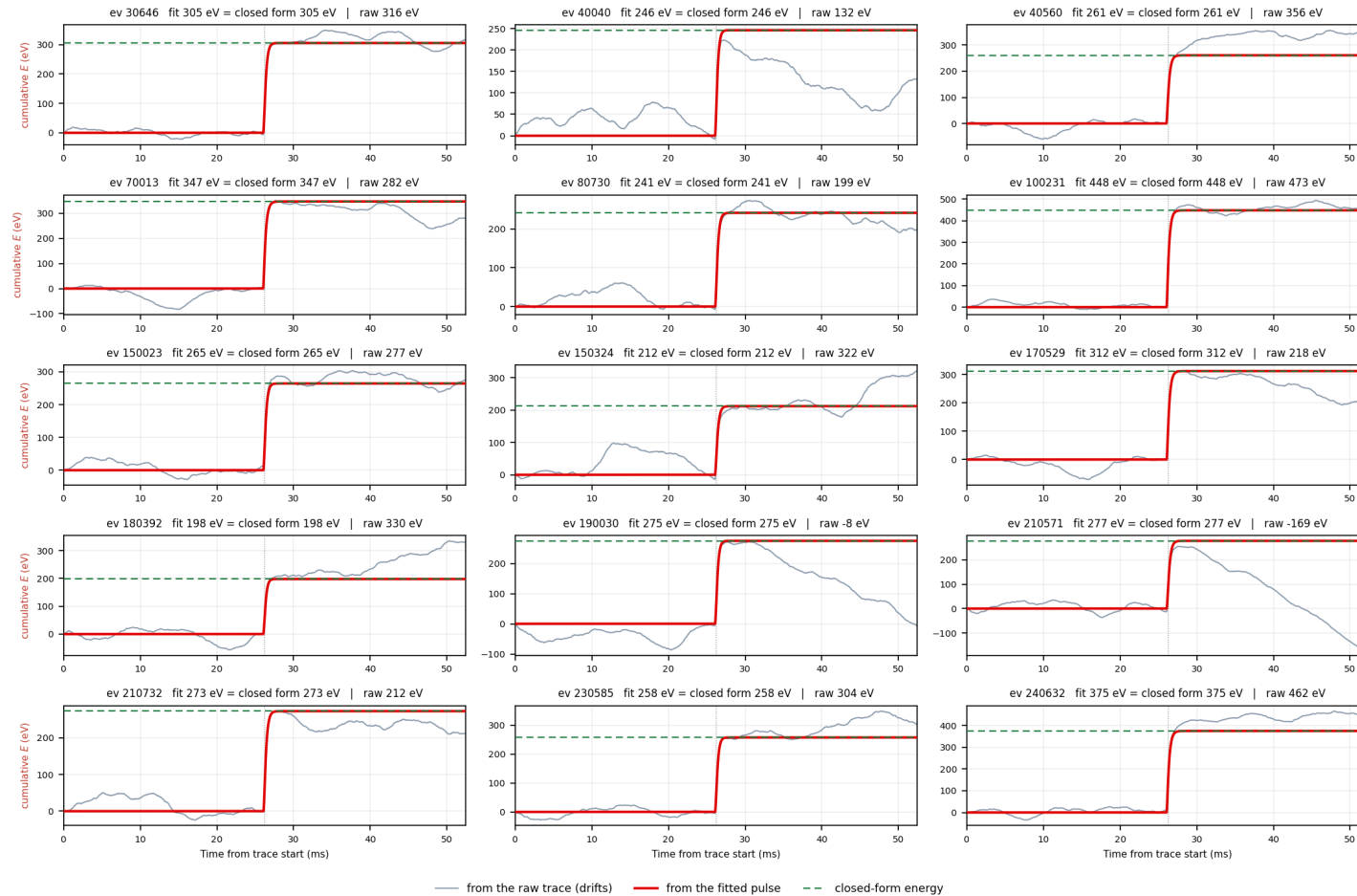
Backup — 15 events, fitted current and power (PBS1)

Z7 PBS1: fitted current pulse (dashed navy) and the power pulse it implies (red)
Z7 24260617_063934, K-line events | method1: $P(t) = I_0(R/L-R_0) \cdot dI(t) + 2 \cdot R \cdot L \cdot (dI(t))^2$ | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
the current is the two-exponential FIT, refitted directly in amperes (model $A[\exp(-t/\tau_r) - \exp(-t/\tau_f)]$), 5 free params, pretrigger free in 16383+-3000; the raw trace is drawn only for reference
integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



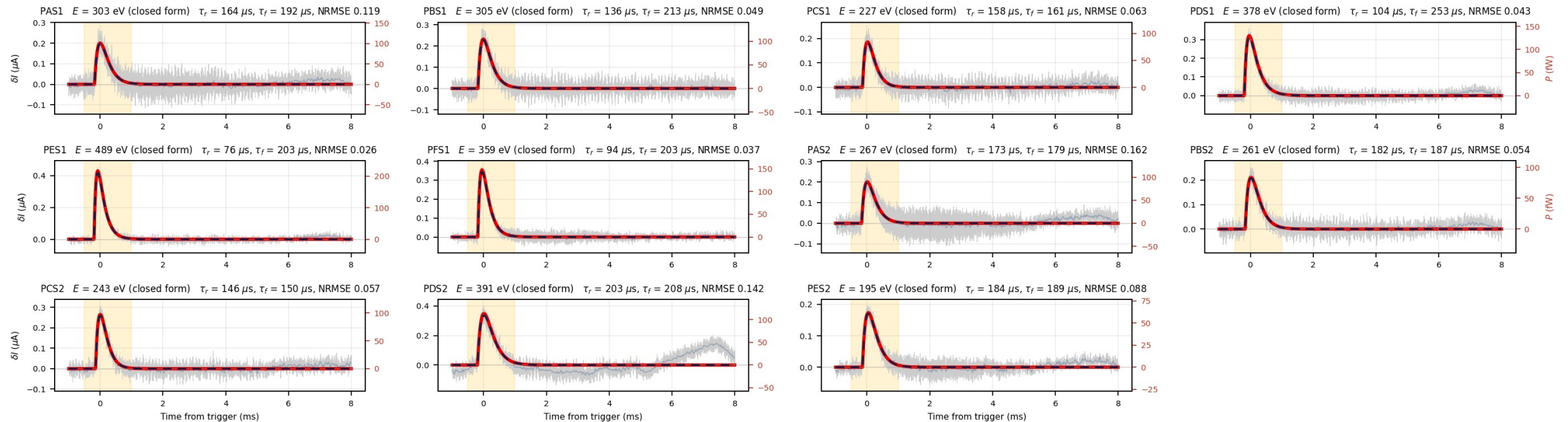
Backup — 15 events, cumulative energy (PBS1)

Z7 PBS1: integrated power = absorbed energy, fitted pulse vs raw trace
 Z7 24260617_063934, K-line events | method1: $P(t) = 10(R_L R_0) \cdot d(t) + 2 \cdot R_L \cdot L \cdot (d(t))^2$ | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
 the current is the two-exponential FIT, refitted directly in amperes (model $A[\exp(-t/\tau_f) - \exp(-t/\tau_r)]$), 5 free params, pretrigger free in 16383+-3000; the raw trace is drawn only for reference
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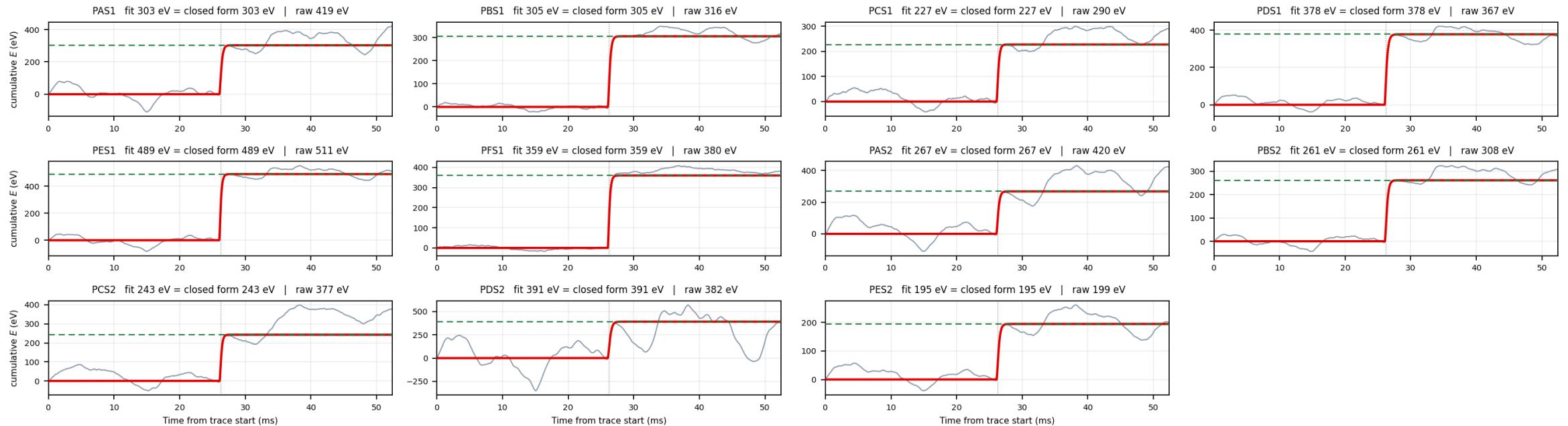
Backup — event 30646, all channels, current and power

Z7 event 30646, all channels: fitted current and power
sum over channels from the fits: 3419 eV = 33.0% of 10.37 keV (official-window raw sum 3397 eV = 32.8%)
Z7 24260617_063934, K-line events | method1: $P(t) = I_0(R_L - R_0) * dl(t) + 2 * R_L * L * (dl(t))^2$ | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
the current is the two-exponential FIT, refitted directly in amperes (model $A[\exp(-t/\tau_f) - \exp(-t/\tau_r)]$, 5 free params, pretrigger free in 16383+-3000); the raw trace is drawn only for reference
integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



Backup — event 30646, all channels, cumulative energy

Z7 event 30646, all channels: integrated power
 sum over channels from the fits: 3419 eV = 33.0% of 10.37 keV (official-window raw sum 3397 eV = 32.8%)
 Z7 24260617_063934, K-line events | method1: $P(t) = I_0(R_{L-R0}) * dl(t) + 2 * R_{L-R0} * (dl(t))^2$ | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
 the current is the two-exponential FIT, refitted directly in amperes (model $A[\exp(-t/\tau_f) - \exp(-t/\tau_r)]$, 5 free params, pretrigger free in 16383+-3000); the raw trace is drawn only for reference
 integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



Backup — energy distribution of every channel

Z7: energy of every K-line event, one panel per channel (30 series)

each entry is one event: its trace fitted with the two-exponential and the power integrated in closed form, method1 quadratic

the dashed curve is a Gaussian fit to the core, so its centre is the PEAK; these distributions are right skewed, so peaks must not be added (the plain means do add, to 3710 eV = 35.8% of the 10.37 keV K line)

